

Exam. Code : 107204

Subject Code : 1666

**Bachelor of Computer Application (BCA) 4th Sem.
(Batch 2021-24)**

DATA STRUCTURES AND FILE PROCESSING

Paper-I

Time Allowed—3 Hours] [Maximum Marks—75

Note :— Attempt **FIVE** questions in all, selecting at least **ONE** question from each section. The fifth question may be attempted from any section. All questions carry equal marks.

SECTION—A

1. What are the features of different types of data structures ? Explain the procedure of time and space trade off. 15
2. (a) What are the different operations that can be performed on Stacks ? Explain. 10
(b) Compare the features of Stacks and linked lists. 5

SECTION—B

3. (a) Discuss the operations that can be performed on trees. 5
(b) How depth first search operations are performed on graphs ? Explain. 10

4. Explain the following :

- (a) Role of trees. 5
- (b) Binary search. 10

SECTION—C

5. Discuss the following sorting techniques by taking suitable examples :
(a) Insertion sort. 7.5
(b) Quick sort. 7.5
6. Explain the working of the following sorting techniques :
(a) Heap sort. 7.5
(b) Bubble sort. 7.5

SECTION—D

7. (a) Discuss the role of different types of file organizations. 10
(b) Explain the concept of Master and transaction file. 5
8. Write notes on the following :
(a) Compaction. 7.5
(b) Blocking. 7.5